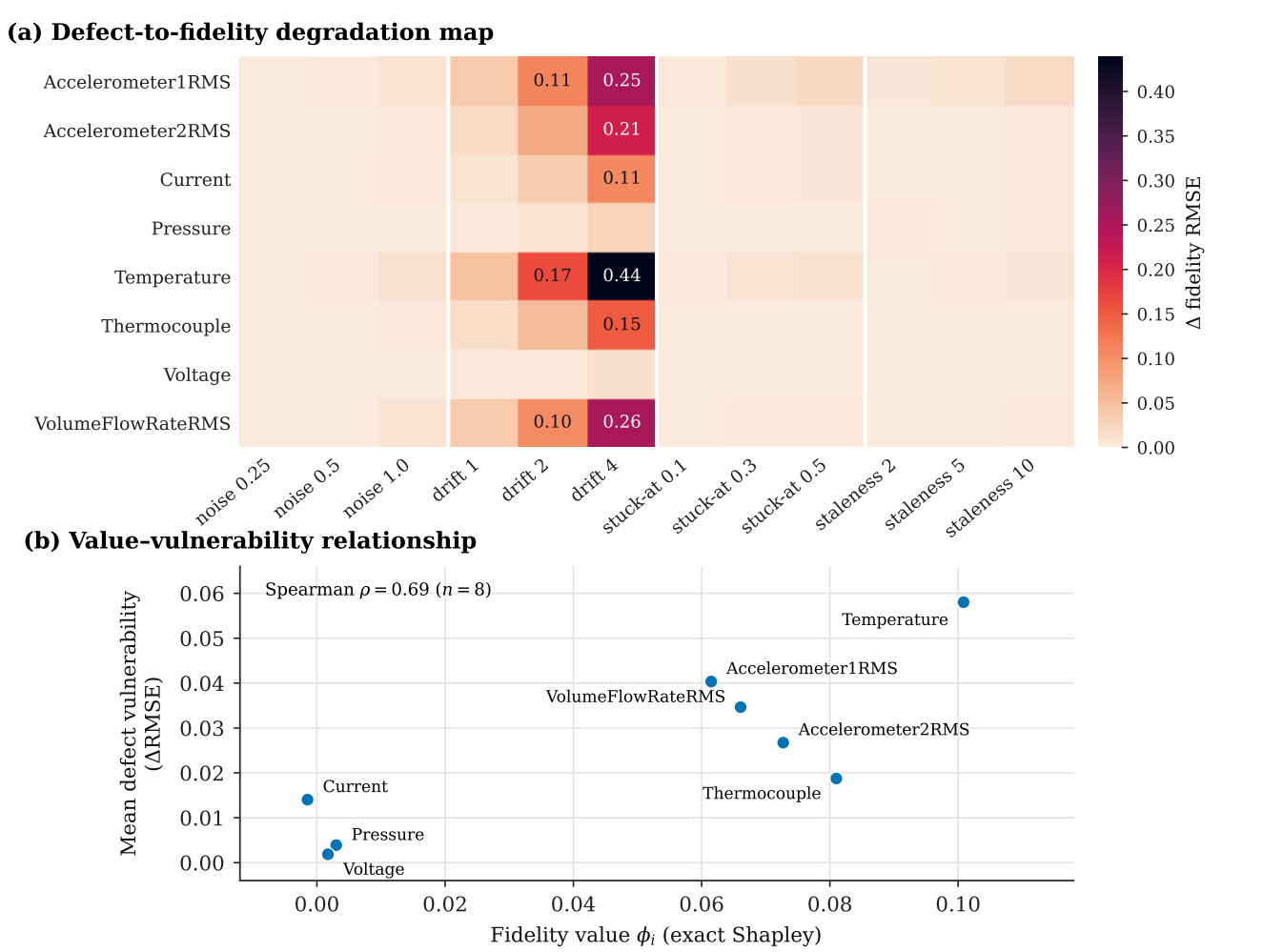
(a) Defect-to-fidelity degradation map
Accelerometer1RMS
Accelerometer2RMS
Current
Pressure
Temperature
Thermocouple
Voltage
VolumeFlowRateRMS
0.11
0.25
0.21
0.11
0.17
0.44
0.15
0.10
0.26
noise 0.25
noise 0.5
noise 1.0
drift 1
drift 2
drift 4
stuck-at 0.1
stuck-at 0.3
stuck-at 0.5
staleness 2
staleness 5
staleness 10
Δ fidelity RMSE
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
(b) Value–vulnerability relationship
Spearman ρ = 0.69 (n = 8)
Temperature
Accelerometer1RMS
VolumeFlowRateRMS
Accelerometer2RMS
Thermocouple
Current
Pressure
Voltage
Mean defect vulnerability (ΔRMSE)
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0.00
0.02
0.04
0.06
0.08
0.10
Fidelity value ϕ_i (exact Shapley)